

The Resolution of Affective Reactivity to Stressful Events

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Repeated assessments in everyday life allow for ecologically valid data on dynamic, within-person stress processes. However, typical designs offer little information on the immediate shape of affective responses following daily stressors, including the influence of situational and person-level variables. In a combined clinical and community sample ($N = 248$; recruited between 2016 and 2018), we employed a high-density intensive-longitudinal protocol (observations $N = 1,442$) to capture the temporal dynamics of affect in response to daily stressful events using a microburst design. Specifically, we implemented an adaptive signal-contingent schedule, where an initial stressor report triggered an intense burst of prompts in 15-min increments over the course of 1 hr inquiring about momentary affect. To model affective microtrajectories, we used multilevel structural equation modeling. A piecewise linear growth model consistently showed the best fit across all indices for both negative and positive affect. Affective responses to momentarily experienced stressors were best captured by a model that allowed for changes in affect trajectories over time (an initial steep decline/increase followed by gradual change), with more stressful situations amplifying these trajectories. Moreover, extraversion significantly influenced the initial rise in positive affect, leading to more pronounced early changes in those with higher levels of extraversion. In contrast, neuroticism had an opposite effect on positive affect, dampening these early changes. Results offer a detailed understanding of daily stress dynamics by providing insights into the immediate and evolving nature of affective responses to stress, with implications for personalized stress management strategies.

Keywords: daily stressors, stress processes, ecological momentary assessment, microtrajectories, microburst design

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People report different affective responses to everyday stressors (Bolger & Schilling, 1991). These differences have been linked to interindividual differences in personality (Leger et al., 2016; Vize et al., 2024). Stress includes both exposure to challenging events and the psychological, behavioral, and neurobiological responses they trigger (Harkness & Monroe, 2016; Smyth et al., 2023). These elements are interrelated and essential for understanding stress outcomes (Monroe, 2008). Studying stress as an individualized, dynamic process is essential, because research shows that stress responses vary not only between but also within individuals (Smyth et al., 2023). Although decades of research have provided extensive insight into daily stress processes and their consequences (e.g., Bolger et al., 1989), there is still limited guidance on how to effectively characterize the dynamic unfolding of stress responses within a day—or at even

more fine-grained temporal resolutions—in response to everyday stressors, such as daily hassles, which are typically minor and recurrent but can nonetheless have meaningful cumulative effects (Wright et al., 2020).

To better understand the nature of daily short-term stress processes, as well as the variability in emotional responses within and between individuals, we used an adaptive high-density sampling method, or microbursts. This approach allowed us to capture stress responses in close temporal proximity after stressful events occur (i.e., reactivity and resolution), assessing their effects on regulatory processes as they unfold in a person's natural environment, while minimizing timing-related bias.

Extensive research has focused on measuring stress responses in daily life. Ecological momentary assessment (EMA) is widely used

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to capture momentary experiences and model intraindividual dynamic processes (Smyth et al., 2023). EMA data have high ecological validity, reducing recall and reporting biases, allowing detection of subtle dynamic changes in psychological processes in real-life contexts (Trull & Ebner-Priemer, 2013). To accurately capture the onset, course, and duration of stress-related responses, however, EMA study designs must match the temporal dynamics of the processes being studied (Hopwood et al., 2022). For numerous psychological constructs, including distinct stress responses, there is limited understanding of their fine-grained temporal patterns. Additionally, observed patterns of change often vary depending on the type of analysis and sampling rate used (Wright & Zimmermann, 2019). Without an adequate assessment design, complex patterns of change, such as nonlinear or discontinuous trajectories representing different stages of the stress process (e.g., reactivity and recovery), may go unobserved, missing opportunities to accurately capture these change processes (Hopwood et al., 2022). Only by adapting the sampling rate to the theorized timeframe of how stress responses unfold can discontinuities or nonlinear patterns be detected. Misaligned design and data can lead to various difficulties. If assessment intervals are too long, they might overlook natural cycles, miss significant events or processes, or heighten the risk of biased retrospection. Intervals that are too short, however, can overburden participants (Ebner-Priemer & Sawitzki, 2007). In physiological studies, the temporal dynamics of processes are often well-understood, and sampling rates are adjusted accordingly. For instance, slower processes like skin temperature are recorded at lower frequencies (4 Hz), while faster processes, such as the electrical activity of the heart via electrocardiograms, are captured at rates up to 2,000 Hz (Ebner-Priemer & Sawitzki, 2007). In psychology, however, few studies have systematically explored the consequences of methodological choices related to timing, and researchers often do not justify their time-based design choices (Kaurin et al., 2023), with only about 17% of EMA studies in psychopathology research reporting a rationale for their rate of sampling (Trull & Ebner-Priemer, 2020).

While sparse, evidence suggests that shorter sampling intervals (e.g., every 15 min, 30 min) are best for processes related to emotion, whereas data from lower frequency sampling rates (e.g., every 2–4 hr) can fail to accurately capture the temporal dynamics among relevant variables and can even resemble random data (Ebner-Priemer & Sawitzki, 2007). Although there is a general recommendation to use shorter time intervals to study temporal dynamics of affective processes (Ebner-Priemer & Sawitzki, 2007), previous studies have employed widely varying timescales ranging from a few hours to more than a day (Janssens et al., 2018). These studies typically use signal- and event-contingent data collection methods and employ a variety of approaches to examine the relations between stress and affect. For example, some studies have examined how negative affect leads to interpersonal stress at the subsequent momentary assessment (e.g., Vize et al., 2024), while others aggregate affect ratings over the course of a day to examine how days with higher reported stress relate to the overall reported affect for that day (e.g., Mroczek & Almeida, 2004). Although these designs can provide valuable information about relations between stress and affect, the sampling frames are too broad to provide insight into fine-grained affective dynamics that are known to occur immediately following a daily stressor. Additionally, typical EMA designs generally treat stressors as

random and interchangeable, overlooking the detailed timeline provided by intensive longitudinal data (Sliwinski & Scott, 2014).

Accurate characterization of the immediate affective response to daily stress requires assessment designs that precisely link daily stressors to their corresponding stress responses, allowing for variability and comparison both between and within individuals. For example, a recent study used a burst design where instances of nonsuicidal self-injury (NSSI) triggered follow-up assessments of negative affect and tension at 10, 20, and 30 min postevent (Störkel et al., 2023). These ratings were compared with those following instances when participants reported that they resisted self-harm urges. The findings indicated an immediate linear decrease in negative affect and tension after NSSI, while a quadratic pattern emerged in response to high-urge events with no NSSI. Importantly, the trajectories of negative affect and tension surrounding NSSI events were highly heterogeneous, suggesting that the impact of NSSI on affective processes varied significantly both across individuals and within the same individual across different events.

Affective responses to daily stressors are likely to show similar patterns of within- and between-person variability. Several factors may influence how the same individual responds to daily stressors at different points in time, including habituation (or lack thereof) to repeated stressors, the time of day they occur, and that some daily stressors will be perceived as more frustrating than others (Dunn & Taylor, 2014; Schommer et al., 2003). While some research has examined how a stressor at one time affects subsequent negative affect (Scott et al., 2019), this area remains relatively underexplored. Most studies assume that these stress dynamics are consistent over time for an individual, with only a few considering exceptions like stationarity, where the process governing stress responses remains stable over time (Bringmann et al., 2017). Additionally, past research often compared average stress responses without accounting for temporal trends in the data, potentially overlooking important within-person variations and the dynamic nature of stress responses over time.

There is also substantial evidence for between-person variability in stress responses, and these between-person differences are crucial for understanding health outcomes. The experience of depressive and borderline personality disorder symptoms, as well as high levels of neuroticism, is associated with more variable and unstable negative emotions, but also more persistently elevated negative emotions (Houben et al., 2015; Suls et al., 1998). Neuroticism influences emotional reactivity in response to daily stress events, with individuals high in neuroticism struggling to return to baseline emotional states (Vize et al., 2024). Particularly prolonged negative affect is a risk factor for psychopathology and is reflected by some individuals having stronger autocorrelation effects for negative affect compared to others (i.e., affective inertia; Kuppens et al., 2012) or sustained selective attention to negative information (e.g., MacLeod et al., 2002). Conversely, the ability to sustain positive emotion is vital for daily functioning, well-being, and health (Pressman & Cohen, 2005). Individuals high in extraversion engage in various activities to restore energy and positive states depending on their daily workload, with different activities and levels of recovery occurring each day (e.g., Sonnentag & Niessen, 2008). Existing evidence suggests that neuroticism and extraversion are particularly relevant to general affective functioning but also to affective responses to stress. In turn, they may also be important to the immediate affective response to daily stressors.

The Present Study

We propose addressing the issues of sampling rate and confounded stressful events and stress responses with a *microburst sampling approach*. This involves adaptive signal-contingent scheduling, where an initial stressor report is followed by brief, high-density surveys assessing affect over the course of 1 hr in 15-min increments. This method allows for direct quantitative articulation of affective trajectories following daily stress, which unfold rapidly following a stress event. Our sampling schedule was informed by empirical considerations. For example, in a retrospective study, Verduyn et al. (2009)¹ found that approximately 80% of discrete emotional episodes resolved within the first hour, with the most pronounced affective shifts occurring in the first 30–60 min. Median durations were brief—16 min for fear, 22 min for anger, and 26 min for joy—and were significantly shaped by the initial intensity and subjective relevance of the experience. These findings underscore the value of short prospective sampling intervals for detecting moment-to-moment affective dynamics in daily life. They are consistent with recommendations suggesting that frequent assessments (e.g., every 15–30 min) are best suited to capturing emotion-related processes, whereas lower frequency sampling (e.g., every 2–4 hr) may fail to adequately reflect temporal patterns and, in some cases, yield data that resemble random noise (Ebner-Priemer & Sawitzki, 2007). Beyond theoretical and empirical justification, our design was also shaped by practical considerations, particularly the need to balance scientific precision with participant burden. Based on insights from previous ambulatory assessment studies (Ebner-Priemer & Sawitzki, 2007), we found that administering four assessments at 15-min intervals over the course of 1 hr struck an effective balance between capturing fine-grained emotional trajectories and maintaining participant engagement.

We aimed to determine the temporal dynamics of the stress response by identifying the optimal form of the affective reaction trajectory over time, assess within-person variability in identified microtrajectories via situational moderators (e.g., ratings of how stressful the event was), and examine how individual differences in personality are linked to the shape of these trajectories. We expect that the typical response will show a downward slope (i.e., a decreasing trajectory) of negative emotions following a stressor and an upward slope for positive affect (i.e., affective reactivity to the stressor). Additionally, we hypothesize that these slopes will be moderated by the affective traits of neuroticism and extraversion, such that neuroticism will be associated with shallower, more gradual, or slower changes in both negative and positive affect following a stressor, indicating slower recovery, and steeper, more pronounced, or faster changes for those high in extraversion, indicating quicker recovery.

Method

Sample

All study procedures were approved by the institutional review board of the University of Pittsburgh (IRB Protocol No. 15120131). A total of 311 community members were recruited between 2016 and 2018, both online and through posted flyers, for a study of personality, daily stress, and social interactions. Of the sample, 64.31% ($n = 191$) had a lifetime history of mental health treatment, 61.78% ($n = 118$) of whom were currently receiving treatment at baseline. Of the participants, 4.5% ($n = 14$) were excluded for failing

to complete a minimum of 10 randomly prompted surveys during the ambulatory assessment protocol, resulting in a final n of 297. Of those 297, $n = 248$ also contributed data to the adaptive microburst component of the study. All participants were between the ages of 18 and 40 and were not currently receiving treatment for a psychosis or a psychotic disorder. The subsample ($n = 248$) ranged in age from 18 to 40 ($M = 28.21$, $SD = 6.44$) and was 55.64% female ($n = 138$). Most participants identified as White (79.03%, $n = 196$). Of the participants, 11.69% identified as Black or African American ($n = 29$), 7.66% as Asian ($n = 19$), <1% as Native American or Alaskan Native ($n = 2$), and 5.65% as “other” ($n = 14$). Two individuals declined to answer this item.

Procedure

Participants completed an ambulatory assessment protocol, where they received six² surveys per day during an approximately 12-hr time window that corresponded to their typical waking hours. The length of the adjusted 14-day window was selected as a pragmatic compromise—sufficient to capture meaningful within-person variability while minimizing participant fatigue and maximizing adherence.

Blocked random intervals were set so that a minimum of 90 min passed between surveys, and participants were given 20 min to initiate a response to each one.

A detailed overview of this *microburst protocol* can be found in Figure 1. If participants endorsed experiencing a stressful or unpleasant event³ within the previous 45 min at a random prompt, this response triggered follow-up microsurveys in 15-min intervals until 1 hr poststress. Depending on when the stressor occurred, participants received differential sets of items to assess their short-term stress response. Participants received up to three kinds of different prompts: retrospective ratings, signal-contingent prompts, and microprompts in 15-min increments over the course of 1 hr. Specifically, if a participant endorsed that a stressful or unpleasant event “just happened,” they would receive four microsurveys at 15 min, 30 min, 45 min, and 1 hr after submitting the random prompt response in addition to the signal-contingent prompt assessing their momentary affective state. If they indicated that the event occurred 30 min prior to the random survey, the participant would receive two microsurveys—45 min and 1 hr poststress event (15 and 30 min postrandom survey)—in addition to the signal-contingent prompt and one retrospective item on how they felt during the stressful event. Participants were given 10 min to respond to each microsurvey. To

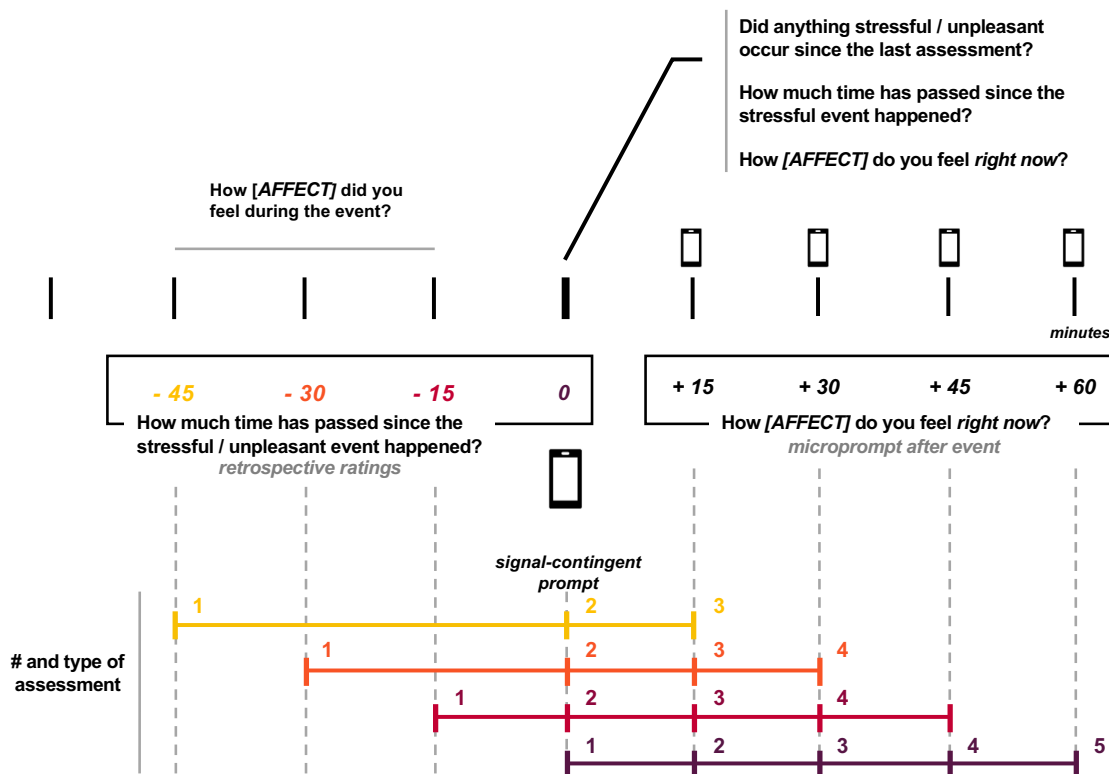
¹ Verduyn et al. (2009) conducted two daily diary studies in which participants retrospectively reported the duration and characteristics of daily emotional episodes (e.g., anger, joy, fear), as well as relevant contextual and personality variables. However, given the inherent limitations of retrospective self-reporting in capturing precise temporal dynamics, the findings remain suggestive and underscore the need for prospective methodologies (see, e.g., Lucas et al., 2021).

² Due to software setup errors, each of the following procedures deviated somewhat for a handful of participants each.

³ In an effort to avoid confounding between the experience of a stressor and the appraisal of the event as stressful, participants were first asked if a “stressful event” occurred and were then given a list of daily hassles to choose from. If they did not endorse any stressful event, they were asked if any of the same list of events had occurred but not labeled as stressful. Thus, the events could be endorsed whether or not the participant would have endorsed them as stressful, and both were used as triggering events here.

Figure 1

Schematic Overview of the Adaptive Microburst Protocol Triggered by Reports of Stressful or Unpleasant Events



Note. When participants reported a stressful event within the past 45 min, follow-up microsurveys were initiated at 15-min intervals for up to 1 h poststress. Depending on the timing of the reported stressor, participants received a combination of retrospective ratings, signal-contingent prompts, and additional microburst assessments. For example, a stressor that “just happened” triggered four surveys over the next hour, while a stressor reported 30 min earlier triggered two surveys. See the online article for the color version of this figure.

reduce participant burden, responses to microprompts were not required for full compensation but were incentivized with additional entries into drawings for an iPad Mini. A total of 20,380 responses to random prompts were collected over the course of the study, with a mean number of 68.62 ($SD = 20.94$) surveys completed per participant. A total of 3,496 microresponses were collected with an average of 14.10 ($SD = 15.79$) per participant, ranging from 0 to 120 microsurveys completed. The median number of microburst assessments completed by each participant was nine.

Measures

Baseline Questionnaires

Participants completed a demographic questionnaire including items addressing sex, age, ethnicity, race, and prior or present mental health treatment.

International Personality Item Pool–NEO. To assess neuroticism and extraversion, we used the International Personality Item Pool–Neuroticism, Extraversion, Openness (NEO; Johnson, 2014), a short-form, open-source alternative to the Revised NEO Personality Inventory (Costa & McCrae, 1992). The International Personality Item Pool–NEO consists of 120 items from the

International Personality Item Pool (Goldberg, 1999) that measure five NEO domains (Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism) and thirty subdomains (six per domain). Items (e.g., “I get angry easily”) are rated on a 5-point Likert scale from 0 (*very inaccurate*) to 4 (*very accurate*). In this study, domain-level omega reliabilities were .88 for Extraversion and .91 for Neuroticism.

EMA

Affect. Six affect items were used in the microsurveys with two adjectives for positive affect (e.g., Confident, Happy) and with four adjectives for negative affect (e.g., Angry, Sad, Nervous, and Ashamed). Depending on the timing across our microburst design (Figure 1), participants were asked to indicate either their momentary affect (e.g., “How ADJECTIVE do you feel right now?”) or their affective experience during the stressful or unpleasant event (e.g., “How ADJECTIVE did you feel during this [stressful] event?”) on a slider scale from 0 (*not at all*) to 100 (*extremely*). McDonald’s ω for the negative affect composite were $\omega_{\text{within}} = .68$ and $\omega_{\text{between}} = .87$. For the positive affect composite, which included only two items, we report within- and between-person interitem correlations ($r_{\text{within}} = .60$; $r_{\text{between}} = .76$).

Stressful Events. Participants were asked whether anything “stressful” had occurred since the last assessment. Participants who reported experiencing a stressful event since the previous assessment could select multiple types of stressors using a “check all that apply” format. The survey included categories such as interpersonal conflict, social disappointment, work- or school-related difficulties, home-related issues, health- or accident-related events, and events affecting others, among others.⁴ These were assessed with items like “Which of the following types of stressors have you experienced since the last assessment?,” followed by specific questions for each selected category (e.g., to differentiate types of conflict, health concerns, or social disappointments).

To streamline subsequent data collection and reduce participant burden, respondents were then asked to identify the single most stressful event among those they had selected (item: “Out of all the stressors you selected, which one is MOST stressful?”). All follow-up questions—regarding emotional reactions, timing, social context, and cognitive processing (e.g., stress intensity, interaction partner, and rumination)—were then anchored to that primary event. This structure was designed to capture the multidimensional nature of daily stress while ensuring consistent and interpretable follow-up data.

Stressfulness of the Event. Participants also rated how stressful or unpleasant the event was at the time it occurred, using a scale from 0 (*not at all*) to 100 (*extremely*).

Participants who indicated that they had not experienced a stressful event since the last assessment were asked if an “unpleasant” event had occurred since the last assessment. They were then asked the type of event, with the same choices offered except the other event was referred to as an “other unpleasant event.” Events were rated on their level of unpleasantness using a scale from 0 (*not at all*) to 100 (*extremely*). These selections were presented in this manner to differentiate situation characteristics from the perception of stress. For this article, ratings of stressful and unpleasant events were combined into one variable, with the purpose of capturing stress events that are less likely to be confounded with perceptions of stress.

Transparency and Openness

All available code to replicate our findings is publicly available via <https://osf.io/h7pvd>. Our analyses were not preregistered because this data set was designed as a resource to explore a wide range of questions with varying effect sizes, rather than to test a single specific hypothesis. Most planned analyses were expected to rely on covariance/correlation matrices (e.g., structural equation modeling and multilevel modeling). Consequently, sample size selection was guided primarily by the goal of achieving stable effect estimates, rather than focusing on the power to detect a specific effect size in the population. Additionally, we aimed to detect small effects consistent with the average effect size reported in personality and social psychology literature. Recent research suggests that correlations of this magnitude stabilize when sample sizes approach $N = 250$ (Schönbrodt & Perugini, 2013). Therefore, for our most conservative analyses—such as between-person associations in this hierarchical data set (bursts and events nested within persons)—we targeted a minimum sample size of approximately $N = 250$.

Data Analytic Approach

Our analyses proceeded in two steps: First, we used multilevel latent growth curve models to identify the optimal functional form of

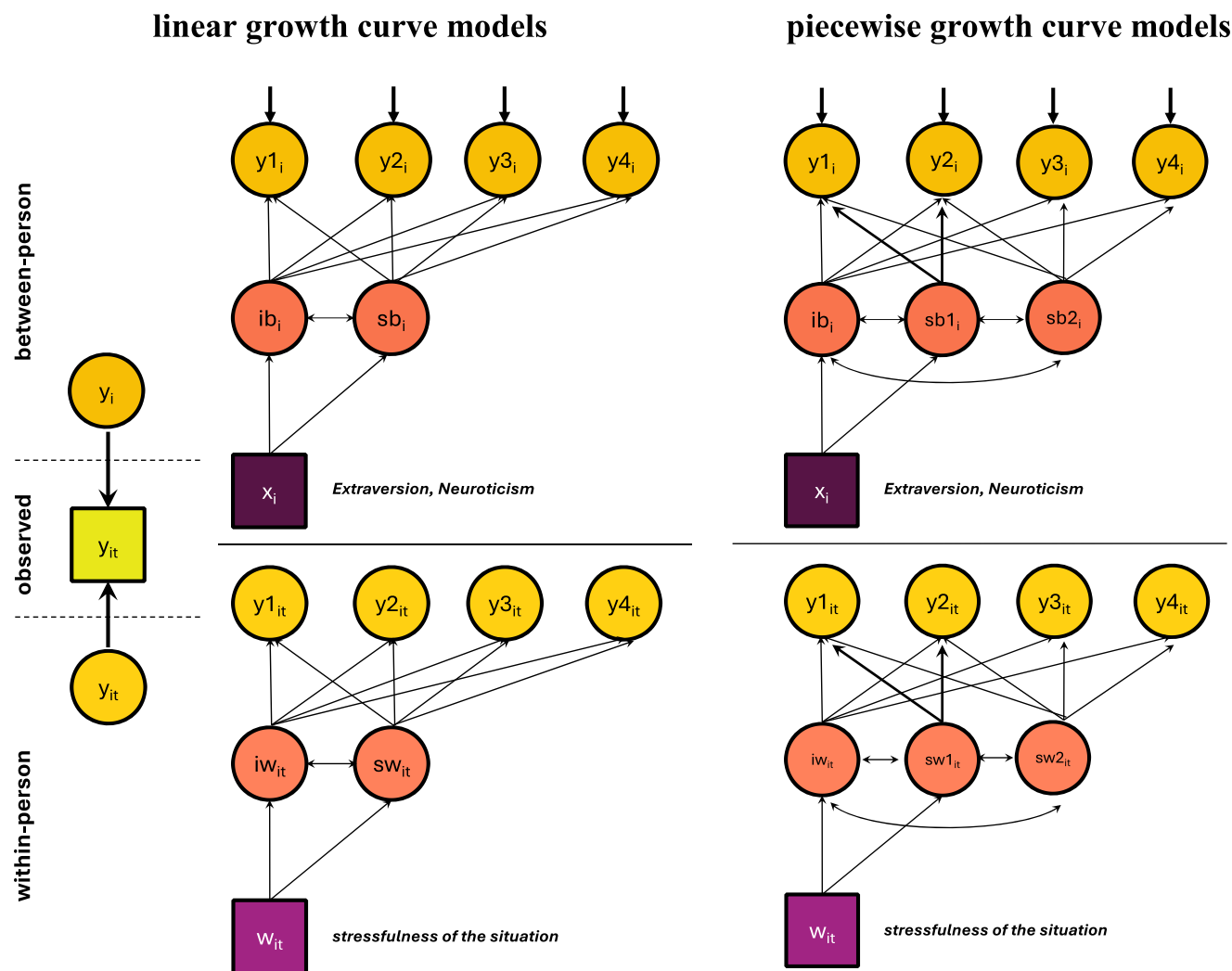
affective microtrajectories over time. In the second step, we included both time-varying predictors (perceived stressfulness/unpleasantness of the event) and time-invariant predictors (baseline Extraversion and Neuroticism) to assess their influence on the random effects of the latent curve models (i.e., random intercepts and slopes). Individual negative and positive affect items were combined into composite negative and positive affect variables for our analyses.

Repeatedly assessing affect over the course of 1 hr creates a hierarchical data structure, with events and their associated series of microburst follow-up assessments (within-person level; Level 1) nested within individuals (between-person level; Level 2). To accommodate the nested data structure and decompose the total variance into latent between-person variance and within-person residual variance, we employed multilevel structural equation modeling (Sadikaj et al., 2021). Multilevel structural equation modeling allowed us to leverage the relative strengths of both multilevel modeling (e.g., decomposition of within- and between-person variance) and structural equation modeling (e.g., flexibility in incorporating predictors and outcomes and handling of missing data) to effectively explore within- and between-person differences in daily stress responses. Specifically, Figure 2 provides a schematic overview of the continuous growth curve and piecewise growth curve models for the present study. Across models, the between-person level of the latent growth curves captures the average or normative affective reactivity response to stress (i.e., latent means) and individual differences (i.e., latent variances) in those responses, whereas the within-person latent growth curves capture event-to-event variation (i.e., latent variances) in trajectories of stress reactivity across participants and individual stressful events. We describe the analytic approach for each separate growth curve model below.

To accurately determine the functional form of positive and negative affect tied to stress events and adequately model the resolution of affective reactivity, we gradually increased the complexity of the growth models, starting with a linear growth factor to test linear change, though based on the plots this seemed implausible. To test whether accelerating (or decelerating in the case of positive affect) rates of change best reflected affective trajectories, we estimated two different models. We first estimated an additional quadratic factor and then estimated exponential curves to model changes in negative affect and logarithmic curves (i.e., the inverse of an exponential curve) to model changes in positive affect. Finally, to allow for a more flexible trajectory and capture more abrupt patterns of change across distinct intervals of microburst assessments, we also estimated piecewise linear models. In the piecewise models, the first linear trajectory was the degree of change in affect from the moment of the stress event to 15 min after representing initial reactivity, while the second trajectory included the subsequent follow-up assessments, representing more sustained processes of affective resolution.

The loadings on the latent slope and intercept for the growth curve models were centered on the initial affect rating in all models (i.e., the affect rating at the time of the stress event). The models also included covariances between the intercept and slope(s). At the between-person level, we applied a similar structure, using latent intercept and slope(s) with the same loading pattern as within individuals. To ensure that all between-person variability was captured by the latent

⁴ Additional detail on how events were categorized is described in the Supplemental Material and outlined in Supplemental Tables S2 and S3 of Vize et al. (2024; available at <https://osf.io/j42h6>).

Figure 2*Schematic Overview of the Tested Two-Level Growth Models*

Note. $y1-4$ = affect assessment at 15-min increments after the stressful event; i_w/i_b = intercept (within-/between-person level); sw = slope parameter at the within-person level; sb = slope parameter at the between-person level; $s1$ = slope characterizing the first segment of the microtrajectory growth curve; $s2$ = slope characterizing the second segment of the microtrajectory growth curve. See the online article for the color version of this figure.

constructs, we fixed the variances of the observed variables at zero. Additionally, we estimated covariances among the intercept and slope(s) to account for potential correlations in growth trends across individuals.

In the second step of our analyses, we regressed the random intercepts and slopes of the between-person growth models on Neuroticism and Extraversion. This allowed us to test whether individual differences in starting values and growth trajectories (i.e., reactivity) could be partially accounted for by these traits. To examine whether the perceived stressfulness of daily stress events impacted within-person random intercepts and slopes, stress ratings of each event were included as time-varying predictors in the models.

Absolute fit was assessed using the χ^2 test, which yields lower values for better fitting models. Given the limitations of the χ^2 test

(i.e., a very stringent test of perfect fit and sensitivity to very small departures from perfect fit in large samples), fit indices were also examined: root-mean-square error of approximation (RMSEA), standardized root-mean-square residual (SRMR, within and between), comparative fit index (CFI), and Tucker–Lewis index (TLI). The CFI and RMSEA offer measures of overall model fit across both levels, while the SRMR provides fit indices specific to each level. RMSEA values $\leq .05$, SRMR $< .08$, and CFI and TLI $\geq .95$ indicate good fit (Hu & Bentler, 1999). To compare nonnested models, we used the Bayesian information criterion, for which lower values indicate better fit, and differences in values of >10 indicate much stronger support for the model with the lower Bayesian information criterion (Raftery, 1995).

All models were estimated in Mplus using a maximum likelihood with robust standard errors estimator (Version 8.4;

Muthén & Muthén, 2019). Missing data were assumed to be missing at random and accommodated using a full information maximum likelihood approach that used all available data in estimation. Significance for all model parameters was based on 95% confidence intervals, with confidence intervals that excluded zero being indicative of a parameter that differed significantly from zero.

Results

Descriptive Information

As previously noted, microburst assessments were only delivered when participants reported that a stressful or unpleasant event had happened 45 min ago or less (e.g., if a participant reported a stressful event took place approximately 45 min ago, they would receive one microburst prompt 15 min later). The total number of stressful or unpleasant events reported during the EMA protocol that had occurred 45 min ago or less was 1,969, with participants reporting 7.94 stressful or unpleasant events on average.⁵ Most of these events were reported as stressful events ($n = 1,940$) compared to unpleasant events ($n = 29$).⁶ Of the 1,969 stressful or unpleasant events, at least one microburst assessment was completed in 1,441 cases (73.2%). We compared the stressfulness ratings of events with microburst data available ($M = 63.91$, $SD = 16.93$) to events without microburst data ($M = 64.40$, $SD = 16.00$), and these ratings were highly similar, suggesting that participants were not more likely to complete microburst assessments based on the perceived stressfulness or unpleasantness of the event.

Regarding the microburst assessments, 44.1% ($n = 635$) were administered after participants reported that the stressor/unpleasant event had “just occurred,” whereas the remaining 55.9% ($n = 805$) reflected retrospective reports of events that had occurred earlier—specifically, 15 min ago ($n = 313$; 21.7%), 30 min ago ($n = 311$; 21.6%), or 45 min ago ($n = 182$; 12.6%). Compliance for the microburst assessments (79.98%) was comparable to compliance for the broader EMA protocol (75.9%).⁷

Identifying the Optimal Functional Form

Evaluation of fit indices for negative affect models indicated that the quadratic model ran into problems with estimation, but the exponential and piecewise models each achieved acceptable fit (see upper half of Table 1). However, the exponential model had a lower Bayesian information criterion (>10), likely due to the increased number of parameters needed for the piecewise model. Moreover, there was significant within- and between-level variability in the trajectories of the observed affect variables. At the within-person level, significant variances were found for both the intercept and slopes, suggesting meaningful individual differences in the initial levels and changes over time. Notably, a significant negative covariance was found between the intercept and both slopes, suggesting that individuals with higher initial reactivity levels of negative affect experienced a quicker initial recovery (i.e., steeper initial slope), but also continued recovery/downward slope compared to a flat line (i.e., a steeper second slope). At the between-person level, significant variability was also observed, particularly for the intercept and the first slope of the piecewise model. This indicates that some participants had higher initial reactivity values

and that participants also differed in the degree of immediate change in negative affect.

A very similar pattern of model fit results was found for microtrajectories of positive affect (see bottom half of Table 1). The piecewise model showed the best fit. Again, a significant negative covariance between the intercept and the first slope indicated that lower initial levels of positive affect (i.e., reactivity) were associated with more rapid increases (i.e., resolution) immediately following a stressor. At the between-person level, significant variability in the intercept and first slope highlighted notable differences in initial levels of positive affect and initial affective response to daily stressors across individuals. Figure 3 displays average trends in microtrajectories of negative and positive affect in the 60 min following the report of a stressful or unpleasant event.

Moderator Analyses

Tables 2 and 3 summarize findings regarding moderator analyses. Please note that while we report both standardized and unstandardized model coefficients, all intercepts and mean values are based on unstandardized estimates. Standardized coefficients are retained only for regression paths and correlations among growth factors, where they aid interpretability. This distinction ensures that observed values are accurately represented and remain consistent with the scale of the raw data.

Situational Moderators

Results related to the perceived stressfulness of the situation are presented in Tables 2 and 3. For negative affect, the model demonstrated good fit ($RMSEA = .047$, $CFI = .981$, $TLI = .977$, $SRMR = .042$ within, $.013$ between). At the within-person level, the perceived stressfulness of the situation had a significant positive effect on initial levels of negative affect ($\beta = 0.51$, $p < .001$) and a negative effect on the first slope of the piecewise model ($\beta = -0.32$, $p < .001$), indicating that more stressful situations were associated with higher initial levels of negative affect at the time of the stress event and less pronounced initial increases. Recall that the intercept was strongly negatively linked with the first slope ($r = -.24$, $p = .002$), meaning higher negative affect during the stress event was related to steeper decreases in negative affect immediately following the daily stressor.

For positive affect, adding perceived stressfulness of the situation as a time-varying predictor resulted in excellent model fit ($RMSEA = .016$, $CFI = .997$, $TLI = .997$, $SRMR = .020$ within, $.010$ between). At the within-person level, the perceived stressfulness of the situation significantly negatively predicted initial positive affect ($\beta = -0.34$, $p < .001$), indicating that higher perceived stressfulness was associated with lower initial levels of positive affect. Perceived stressfulness had

⁵ The total reported number of stressful or unpleasant events was 2,990, meaning that $n = 1,021$ (34%) of reported stressors occurred an hour ago or more. Participants reported stressors/unpleasant events in 17.6% of EMA surveys on average.

⁶ Given this strong imbalance between stressful and unpleasant events, we did not include event type as a covariate.

⁷ For the subset of individuals who agreed to receive microburst assessments ($n = 248$), a total of 22,386 random prompts were initiated in the broader EMA protocol (based on 21 Days $\times$ Six Prompts $\times$ 37 Participants in Study 1 and 14 Days $\times$ Six Prompts $\times$ 211 Participants in Study 2). Of these, 16,986 surveys were completed, yielding an overall compliance rate of 75.9%.

Table 1*Fit Indices for Differential Multilevel Growth Models of Negative and Positive Affect*

Model	RMSEA	SRMR w/b	CFI	TLI	BIC	χ^2
Negative affect						
Linear	.103	.059/.074	.87	.90	40,456.334	424.700
Quadratic						
Exponential	.057	.061/.023	.96	.97	40,180.547	148.912
Piecewise	.049	.048/.013	.98	.98	40,236.753	85.724
Positive affect						
Linear	.139	.106/.107	.72	.78	47,917.079	755.456
Quadratic						
Logarithmic	.045	.049/.034	.97	.97	47,254.744	114.789
Piecewise	.019	.022/.010	.99	.99	47,306.651	28.415

Note. $N_{\text{between}} = 248$; $N_{\text{within}} = 1,442$. Rows presented in bold indicate the best fitting model according to fit indices. No values are provided for the model with quadratic trajectories, because it resulted in a nonpositive definite covariance matrix. To ensure model identification, we applied standard constraints—specifically, either by fixing the residual variance at the first time point to zero or by setting all residual variances to be equal. Given that this constraint was already in place, we believe the estimation issues encountered with the quadratic model were likely due to sampling variability or convergence instability rather than a structural flaw in the model. To address this, we tested two alternative estimation strategies: (a) using the maximum likelihood estimator with standard errors approximated by first-order derivatives and (b) increasing the number of random starts using the STARTS = 20 option in the ANALYSIS section. None of the approaches yielded reliable results. RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual; w/b = within- and between-person level; CFI = comparative fit index; TLI = Tucker–Lewis index; BIC = Bayesian information criterion.

a small negative effect on the initial slope ($\beta = -0.12$, $p = .023$) and a positive effect on the second slope ($\beta = 0.19$, $p = .004$). Recall that the intercept and initial slope were positively correlated ($r = .31$, $p = .030$), while the intercept and second slope were negatively correlated ($r = -.33$, $p = .002$). These findings suggest that individuals who report higher positive affect immediately after a stressor tend to experience a more rapid early increase in positive affect (i.e., quicker short-term recovery), but their affect stabilizes more slowly over time. At the between-person level, relationships between initial levels and growth rates were not significant. When experiencing higher momentary stress, this process is amplified, leading to lower initial positive affect reactivity and a steeper increase in positive affect afterward.

Individual Differences in Personality

In the final step, Extraversion and Neuroticism were added as time-invariant covariates to our two-level growth model. For all models, fit ranged from *good* to *excellent* (value ranges: RMSEA = .016–.047, CFI = .979–.997, TLI = .975–.996, SRMR = .022–.048 within, .010–.012 between). [Supplemental Figure S1](#) illustrates observed trajectories separately for individuals low (-1 SD) and high ($+1$ SD) in Extraversion and Neuroticism.

Higher levels of Extraversion were significantly linked to lower initial levels of negative affect during daily stressors ($\beta = -0.17$, $p = .041$) but did not predict changes over time (associations with both slopes $p > .1$; [Table 2](#)). Extraversion significantly predicted higher initial levels of positive affect ($\beta = 0.37$, $p < .001$) as well as steeper initial increases in positive affect ($\beta = 0.25$, $p = .011$). There was, however, no significant effect of Extraversion on the later segment of the affect trajectory ($\beta = -0.11$, $p = .600$; [Table 3](#)). This pattern suggests that individuals with higher Extraversion experienced smaller initial drops—or maintained higher levels—of positive affect in response to stressors, followed by more rapid early increases in positive affect during the initial recovery period. In contrast, longer term changes in positive affect did not differ by Extraversion.

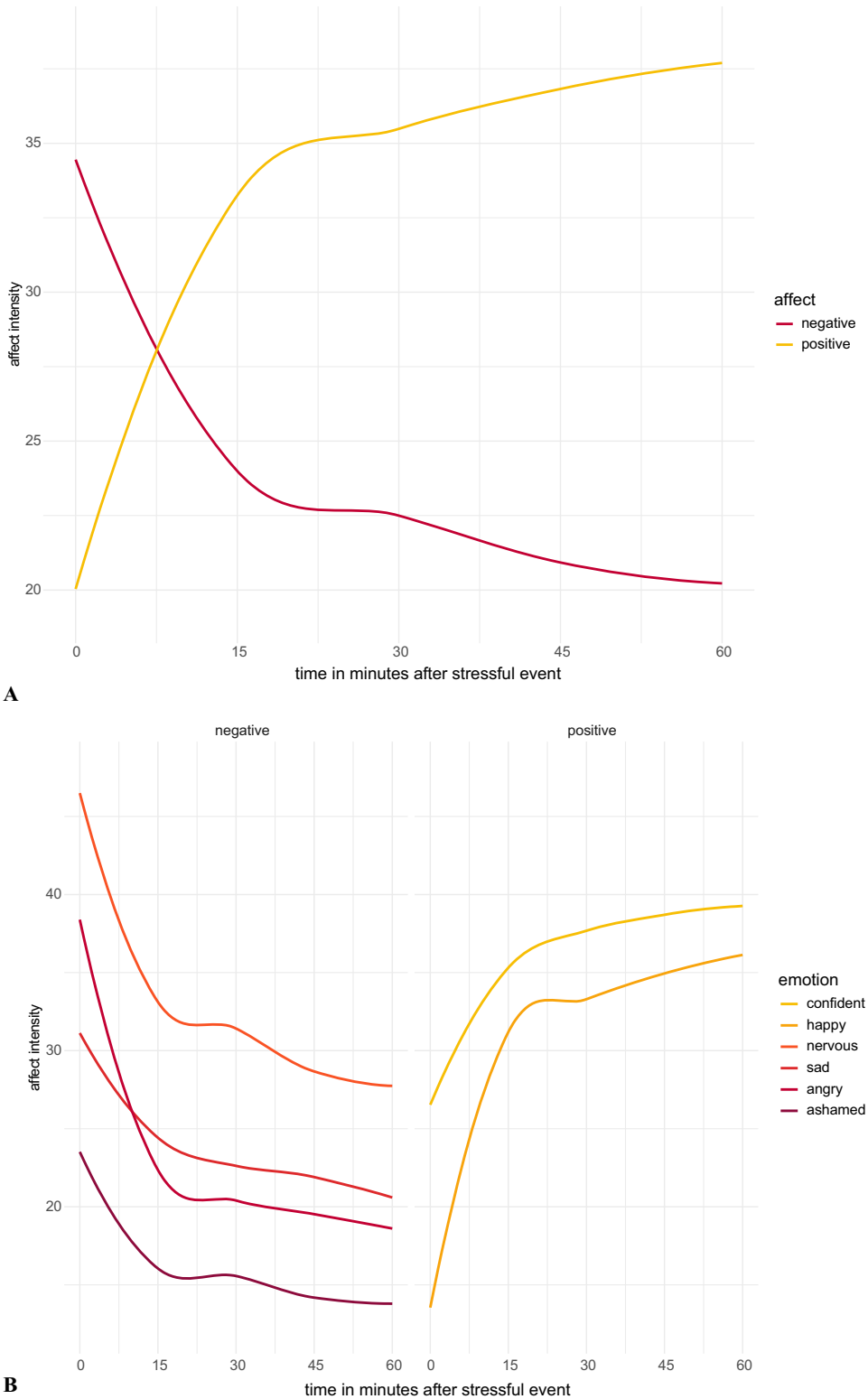
A similar pattern emerged for Neuroticism and negative affect. Higher levels of Neuroticism significantly predicted higher initial levels of negative affect ($\beta = 0.35$, $p < .001$). At the same time, Neuroticism did not significantly affect the change in negative affect over time, as associations with growth factors (slopes) were non-significant ($ps = .24$ and $.79$; [Table 2](#)). Finally, Neuroticism significantly predicted lower initial levels of positive affect ($\beta = -0.41$, $p < .001$) as well as a dampened initial ($\beta = -0.29$, $p = .004$) but not subsequent response ($\beta = -0.08$, $p = .700$) to daily stressors. In summary, while Neuroticism was strongly associated with higher initial reactivity of negative and positive affect, it did not significantly impact how affect evolved over time (i.e., *resolution*; [Table 3](#)).

Discussion

Much of the existing EMA literature has relied on assessments spaced at relatively coarse intervals (typically 1–3 hr), limiting the ability to capture the temporal unfolding of emotional reactivity and regulation in everyday life ([Verduyn et al., 2015](#)). Such designs may miss more rapid changes in affect and can conflate initial emotional responses with subsequent regulation processes. Moreover, recent critiques (e.g., [Dejonckheere et al., 2019](#); [Kalokerinos et al., 2020](#)) have questioned whether commonly used indices of affective dynamics explain variance in emotional well-being above and beyond mean affect levels, casting doubt on their explanatory value.

In contrast, we used high-density sampling following stressful events to closely examine the trajectory of affective responses to stress in daily life. This design provides the necessary temporal resolution to distinguish initial reactivity from early recovery—an advancement over prior work relying on single-time point analyses or designs with larger time gaps between assessments. Our statistical modeling approach enabled us to assess average emotional response patterns over time, individual differences in those trajectories (i.e., between-person variability), variability from event to event within individuals (i.e., within-person variability), and the situational and dispositional factors influencing these patterns.

Figure 3
Observed Microtrajectories of Negative and Positive Affect Following a Stressful or Unpleasant Event



Note. (A) The figure shows the composite scores for overall positive and negative affective states. (B) The figure shows the trajectories of specific positive and negative affective states. Shaded areas represent the confidence intervals. See the online article for the color version of this figure.

Table 2
STD and UNSTD Parameters (Coeff, p Value) for Piecewise Models of Negative Affective Microtrajectories

Negative affect	Model 1		Model 2		Model 3		Model 4	
	STD	UNSTD	STD	UNSTD	STD	UNSTD	STD	UNSTD
Within-person								
sw1 ↔ sw2	−0.033 (0.789)	−0.718 (0.796)	−0.032 (0.807)	−0.647 (0.813)	−0.020 (0.874)	−0.436 (0.877)	−0.033 (0.790)	−0.720 (0.796)
sw1 ↔ Intercept	−0.397 (0.000)	−37.946 (0.000)	−0.241 (0.002)	−17.903 (0.021)	−0.399 (0.000)	−37.575 (0.000)	−0.399 (0.000)	−38.178 (0.000)
sw2 ↔ Intercept	−0.349 (0.000)	−14.530 (0.000)	−0.386 (0.000)	−13.186 (0.000)	−0.356 (0.000)	−14.837 (0.000)	−0.349 (0.000)	−14.514 (0.000)
Intercept on stressfulness			0.514 (0.000)	0.412 (0.000)				
Stressfulness → sw1			−0.318 (0.000)	−0.135 (0.000)				
Stressfulness → sw2			−0.091 (0.060)	−0.017 (0.063)				
Between-person								
sb1 ↔ sb2	0.794 (0.517)	2.016 (0.200)	0.800 (0.495)	2.206 (0.183)	0.620 (0.605)	1.492 (0.362)	0.832 (0.465)	2.148 (0.163)
sb1 ↔ Intercept	−0.264 (0.053)	−19.638 (0.121)	−0.288 (0.024)	−22.138 (0.069)	−0.335 (0.011)	−24.038 (0.058)	−0.303 (0.026)	−21.545 (0.087)
sb2 ↔ Intercept	−0.306 (0.471)	−1.944 (0.422)	−0.335 (0.417)	−2.347 (0.364)	−0.207 (0.631)	−1.138 (0.610)	−0.268 (0.485)	−1.756 (0.478)
Intercept	2.478 (0.000)	33.773 (0.000)	2.429 (0.000)	33.935 (0.000)	1.496 (0.000)	20.444 (0.000)	3.044 (0.000)	41.502 (0.000)
sb1	−1.863 (0.000)	−10.150 (0.000)	−1.888 (0.000)	−10.382 (0.000)	−2.233 (0.000)	−12.677 (0.000)	−1.267 (0.014)	−6.830 (0.003)
sb2	−3.037 (0.428)	−1.416 (0.000)	−2.803 (0.386)	−1.404 (0.000)	−2.941 (0.509)	−1.270 (0.034)	−3.484 (0.354)	−1.733 (0.001)
E → Intercept							−0.165 (0.041)	−3.576 (0.045)
E → sw1							−0.186 (0.120)	−1.587 (0.131)
E → sw2							0.199 (0.582)	0.157 (0.538)
N → Intercept					0.353 (0.000)	7.084 (0.000)		
N → sw1			0.156 (0.240)	1.305 (0.244)		1.305 (0.244)		
N → sw2			−0.103 (0.800)	−0.065 (0.797)		−0.065 (0.797)		

Note. $N_{\text{between}} = 248$; $N_{\text{within}} = 1,442$. Values in bold indicate that the p value surpasses the significance threshold of 0.05. → indicates regression; ↔ indicates covariation. STD = standardized; UNSTD = unstandardized; sw = slope parameter at the within-person level; s1 = slope characterizing the first segment of the microtrajectory growth curve; s2 = slope characterizing the second segment of the microtrajectory growth curve; stressfulness = experienced stressfulness of the situation; sb = slope parameter at the between-person level; E = Extraversion; N = Neuroticism.

Table 3
STD and UNSTD Parameters for Piecewise Models of Positive Affective Microtrajectories

Positive affect	Model 1		Model 2		Model 3		Model 4	
	STD	UNSTD	STD	UNSTD	STD	UNSTD	STD	UNSTD
Within-person								
sw1 ↔ sw2		2.560 (0.518)	0.117 (0.538)	2.660 (0.500)	0.106 (0.578)	2.428 (0.545)	0.101 (0.590)	2.329 (0.560)
sw1 ↔ Intercept		23.309 (0.001)	0.313 (0.030)	20.302 (0.005)	0.336 (0.016)	23.348 (0.001)	0.335 (0.015)	23.400 (0.001)
sw2 ↔ Intercept		-10.557 (0.000)	-0.332 (0.002)	-8.658 (0.002)	-0.373 (0.000)	-10.667 (0.000)	-0.370 (0.000)	-10.568 (0.000)
Intercept on stressfulness								
Stressfulness → sw1			-0.343 (0.000)	-0.192 (0.000)				
Stressfulness → sw2			-0.117 (0.023)	-0.054 (0.024)				
Between-person								
sb1 ↔ sb2			0.180 (0.004)	0.034 (0.004)				
sb1 ↔ Intercept	0.112 (0.553)	1.694 (0.481)	0.240 (0.538)	1.794 (0.450)	0.244 (0.549)	1.745 (0.461)	0.335 (0.454)	2.350 (0.323)
sb2 ↔ Intercept	0.337 (0.011)	7.269 (0.499)	0.092 (0.511)	7.393 (0.502)	-0.029 (0.841)	-1.955 (0.842)	-0.009 (0.950)	-0.626 (0.950)
Intercept	-0.372 (0.000)	-0.968 (0.746)	-0.122 (0.672)	-1.287 (0.667)	-0.133 (0.663)	-1.244 (0.663)	-0.057 (0.846)	-0.532 (0.846)
sb1	2.067 (0.000)	21.642 (0.000)	2.032 (0.000)	21.657 (0.000)	3.216 (0.000)	33.315 (0.000)	0.824 (0.039)	8.518 (0.024)
sb2	1.895 (0.000)	14.280 (0.000)	1.897 (0.000)	14.288 (0.000)	2.688 (0.000)	20.330 (0.000)	1.085 (0.013)	8.096 (0.004)
E → Intercept	1.054 (0.000)	1.078 (0.000)	1.083 (0.077)	1.074 (0.000)	1.336 (0.144)	1.322 (0.067)	1.460 (0.169)	1.424 (0.038)
E → sw1							0.372 (0.000)	6.111 (0.000)
E → sw2							0.246 (0.011)	2.909 (0.016)
N → Intercept							-0.113 (0.600)	-0.175 (0.581)
N → sw1								
N → sw2								

Note. $N_{\text{between}} = 248$; $N_{\text{within}} = 1,442$. Values in bold indicate that the p value surpasses the significance threshold of .05. → indicates regression; ↔ indicates covariation. STD = standardized; UNSTD = unstandardized; sw = slope parameter at the within-person level; sb1 = slope characterizing the first segment of the microtrajectory growth curve; sb2 = slope characterizing the second segment of the microtrajectory growth curve; stressfulness = experienced stressfulness of the situation; sb = slope parameter at the between-person level; E = Extraversion; N = Neuroticism.

The optimal model for negative affect was a piecewise linear model with a breakpoint at the first 15-min follow-up assessment, showing an initial rapid recovery followed by a plateau. Significant individual- and event-level variability was observed, with higher initial negative affect reactivity levels leading to more rapid changes over time. The stressfulness of the situation influenced both the baseline level and the initial response. Additionally, higher extraversion was associated with lower initial decreases in positive affect after stress, as well as faster recovery, whereas higher neuroticism was associated with stronger drops in positive affect as well as slowed the return to baseline. Like recent studies employing burst designs for high-urgency events (Störkel et al., 2023), our study demonstrates that high-density sampling is essential for detecting variability in stress responses, allowing more accurate assessments of how situational factors and personality traits shape differential affective stress processes (i.e., overall resolution).

Building on these findings, it is noteworthy that much of the affective resolution occurs within the first 15 min following a stressor, highlighting this period as a critical window in the stress response. This rapid initial change underscores the importance of early interventions or coping mechanisms during this brief but pivotal time-frame. However, it is equally important to recognize that individuals vary significantly in how they respond to stress—both across situations and in their average stress response patterns (e.g., Sonnentag & Niessen, 2008). These differences emphasize the need for a more nuanced understanding of stress dynamics and tailored approaches that account for both situational and individual variability.

To illustrate, our findings demonstrate that perceived stressfulness significantly moderated affective reactivity at the within-person level. More stressful events elicited sharper affective responses, characterized by heightened negative affect and dampened positive affect at onset, but also faster short-term positive affect recovery. One possible explanation is that stronger emotional responses may elicit more effortful emotion regulation, with high-intensity affect potentially disrupting immediate regulation and requiring more time to settle. At the same time, we acknowledge that the observed pattern may partly reflect statistical phenomena rather than true regulatory success. Notably, while Neuroticism accounted for variation in initial affective levels, it did not predict change in affect over time, suggesting that trait-level affectivity may shape reactivity more than recovery. These findings underscore the importance of interpreting slope patterns in considering features and statistical dependencies and point to the need for future studies to further disentangle dynamic emotion regulation from statistical artifacts. Relatedly, of the microburst assessments, 44.1% reflected prospective reports (“just occurred”), whereas 55.9% reflected retrospective reports (15, 30, or 45 min earlier). Accordingly, the slope for the first spline (Slope1) was informed by a relatively balanced mix of prospective and retrospective accounts, with a slight predominance of the latter. This, too, should be taken into account when interpreting early affective change. The empirical question of how recall type may shape trajectory estimates remains open and likely requires study designs with sufficient power and granularity to examine this issue more definitively.

Notably, between-person differences in stress ratings did not significantly affect growth rates for either affect. Consistent with prior studies, higher neuroticism was associated with stronger negative affective reactivity and slower recovery, indicating affective inertia that may increase psychopathology risk (Houben et al., 2015; Kuppens et al., 2012). In contrast, higher extraversion was linked to more modest initial positive affect reactivity and more rapid recovery,

supporting the idea that more extraverted individuals employ adaptive strategies to sustain positive emotions (Pressman & Cohen, 2005).

Our design captures the complete stress response cycle regarding daily hassles—rapid increase, decrease, and stabilization—through high-resolution sampling, offering greater precision than typical intensive-longitudinal studies. This granularity has important implications for designing effective just-in-time adaptive interventions. Just-in-time support means delivering the right support at the right moment—neither too early nor too late (Nahum-Shani et al., 2018), as timing greatly affects the effectiveness of the intervention. Timing refers to the precise point when a process starts or ends, marking specific conditions at that moment. Our findings show that vulnerability and opportunities for change can emerge quickly, often triggered by stressors, highlighting the need to detect potentially personally relevant high-risk states that demand immediate support. High-density sampling captured distinct reactivity and recovery phases shaped by individual traits and situational factors, suggesting that just-in-time adaptive interventions should adapt to rapid affective shifts, especially during initial negative affect reactivity. The observed individual differences suggest a need for personalized intervention strategies: For instance, highly extraverted individuals may need less support overall, while those high in neuroticism, who tend to recover more slowly, might require extended support. Additionally, the influence of situational stressfulness on baseline and initial responses indicates the importance of adjusting intervention intensity according to stressor severity. Tailoring just-in-time adaptive interventions to both personality and situational factors could enhance their effectiveness in fostering timely emotional regulation and adaptive stress recovery.

We were unable to capture affect ratings immediately before stressful events, which may have caused us to miss the initial change in affect, potentially limiting our ability to fully assess the reactivity effect. This limitation stems from the nature of the process studied, as it is nearly impossible to prompt for data collection immediately before a stressful event occurs. Determining the ideal baseline value for affect before stressor events remains a challenging question and depends on a study’s specific goals and design. Smyth et al. (2023) outlined three potential approaches: the proximal baseline (assessments immediately before a stressor), the local baseline (averaging affect over a recent window, e.g., 24 hr), and the cumulative baseline (aggregating all nonstressor moments across the study). Each comes with trade-offs: The proximal baseline is timely but potentially biased by anticipation, the local baseline is stable but sensitive to unrelated affective events, and the cumulative baseline is robust but less responsive to recent changes. Critically, these strategies have yet to be systematically compared.

The Reactivity of Affect in the Daily Lives of Youth study (see preregistration at <https://osf.io/tsa4v>; Philippi et al., 2025) is specifically designed to address this gap by empirically evaluating how different baseline definitions shape estimates of affective reactivity in daily life. Building on the current design, Reactivity of Affect in the Daily Lives of Youth will directly compare (a) a proximal baseline using the most recent standard EMA prompt without a stressor, (b) a flexible approach using the last completed prompt of any type, and (c) a contextual baseline based on person-mean affect during nonstress moments. By testing these alternatives side-by-side, the Reactivity of Affect in the Daily Lives of Youth study aims to empirically narrow current gaps in the literature by clarifying how baseline definitions influence estimates of emotional change, thereby informing best practices for modeling real-world stress responses—including

more precise operationalizations of key components such as affective reactivity and recovery.

Although our microsampling design enabled a fine-grained tracking of stressor-related affective responses, it did not include a measure of whether stressors were perceived as resolved or ongoing. To better capture the full trajectory of emotional responses to daily stressors, future studies could benefit from a multimethod ambulatory assessment framework that combines event-, interval-, and signal-contingent sampling. In such a design, participants receive randomized prompts six times per day (signal-contingent) and report emotionally relevant events as they occur (event-contingent). Ten minutes after each reported event, they complete a follow-up survey (interval-contingent) to assess short-term emotional responses. Later signal-contingent prompts can determine whether the event remains emotionally relevant, allowing researchers to examine both short- and long-term processes of stress regulation. Such an intensive longitudinal approach may offer more nuanced insight into the use and effectiveness of stress regulation strategies over time. Critically, incorporating additional stressor characteristics—such as perceived controllability—provides important context for interpreting individual differences in emotional reactivity and recovery (Lazarus & Folkman, 1984), as such appraisals influence the choice of emotion regulation strategies, which in turn shape the temporal dynamics of affective responses (Verduyn et al., 2013).

Additionally, we did not employ a control condition, which limits our ability to compare the effects of stressors against a nonstress condition. This makes it difficult to isolate the specific impact of the stressors and fully understand their effects relative to reactivity and regulation to other types of events. Our study design imposed a relatively high participant burden; however, high compliance rates suggest it was generally well-accepted. Participants had to complete six random prompts per day, along with additional follow-up microburst reports for highly stressful moments. This may have led to systematic missing data in the microburst follow-up sequences. Additionally, we cannot rule out the possibility that some participants underreported stressful events to avoid the frequent sampling required after reporting a stressful or unpleasant event. Finally, the inclusion of “confident” as an indicator of positive affect was guided by both practical limitations and theoretical rationale. Because the microburst design constrained the number of affect items we could include, it required us to prioritize adjectives with strong empirical support and conceptual relevance. “Confident” was selected for its association with agency, self-efficacy, and self-regulation—core components of adaptive stress recovery. Future research should broaden the affective scope to include lower arousal states such as “calm” and “content,” which may offer insight into alternative recovery trajectories shaped by context-specific stressors and individual differences in arousal regulation.

Constraints on Generality

The generalizability of our results is limited to young, predominantly White, cisgender women who frequently encountered stress, restricting our findings on affect-regulatory dynamics to this demographic. Our sample lacked diversity to explore daily stressors unique to minority groups. Individuals of other sexes, genders, or ethnic minority groups—who may experience different stress levels and negative affect due to factors like discrimination—might show different patterns. Therefore, our conclusions may not extend to social identity-related stressors, which could have more pronounced effects.

Nonetheless, our multilevel approach provides a framework for future studies on microaggressions.

Conclusion

Using a 15-min interval microburst sampling framework, our design effectively captured the full stress response cycle—initial rapid change and stabilization—while accounting for both between- and within-person variability. This approach improves precision over traditional EMA protocols, which often miss critical moments of the stress response. Our findings show individual differences in reactivity, with a rapid recovery at 15 min followed by a plateau, intensifying with greater stressfulness. Extraversion was linked to a sharper initial increase in positive affect, while Neuroticism had no significant impact on reactivity trajectories. These results highlight how personality and situational factors shape stress responses, informing tailored just-in-time interventions.

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